

IoT-Based Smart Insole with Sensor for Testing Leg Tissue and Skin

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Abstract— This project introduces a novel approach to foot health monitoring through the development and implementation of an IoT-based smart insole with integrated sensors. The primary objective is to provide continuous real-time testing of leg tissue and skin, particularly focusing on parameters such as temperature, pressure distribution, and SpO₂ levels. The project utilizes advanced technologies, including thermal sensors, force-sensitive resistors, and a Max30105 sensor integrated into an Arduino Uno microcontroller. A Wi-Fi and Bluetooth-enabled ESP32 module enables wireless communication, facilitating the seamless transmission of data to a secure cloud platform. The project's key findings reveal the successful integration of sensor technologies into a wearable insole for continuous leg tissue and skin testing. The findings underscore the potential impact of continuous monitoring on improving patient outcomes and highlight the versatility of the Smart Insole in addressing critical health challenges.

Keywords— *Diabetic foot ulcer, smart insole, safety, cloud, IoT, NodeMCU, accessibility, power backup, performance.*

I. INTRODUCTION

The inability of the body to use its own insulin or an inadequate insulin production by the pancreas are the two main causes of diabetes, a chronic disease [1]. Diabetic foot ulcers are a common complication of poorly controlled diabetes mellitus. Inadequate treatment of the feet, peripheral vascular disease, neuropathy, or poorly controlled diabetes are common causes. Amputations of the lower limbs and foot osteomyelitis are both caused by it. Pressure points and areas of the foot that experience repeated trauma are common sites for the development of these ulcers [2]. In order to avoid problems like infections and amputations, it is essential to discover foot health issues early and monitor them continuously. In response to these difficulties, this project will build a system to detect foot ulcers using sensors, microcontrollers, and cloud computing. The prevalence of foot ulcers is a pressing issue in healthcare, with a considerable impact on the quality of life for affected individuals. Current preventive measures and monitoring practices often lack the real-time data necessary for timely intervention. This project proposes a comprehensive solution by integrating various sensors and wireless communication to create a system capable of continuous monitoring and early detection of potential foot ulcers [3].

In the presence of early diagnosis and treatment, diabetes-related complications, such as unwanted amputations and treatment costs, can be effectively avoided. An essential component in the healing of wounds is oxygen. One of the elements (a typical symptom) that prevents diabetic wounds from healing is suboptimal oxygenation, or hypoxia, which reduces the rate of blood flow to the wound [4].

In recent years, new shoe designs and innovative foot sensors have been developed in response to the need for progressive methods to both prevent and reduce foot ulceration among diabetic patients. Termed diabetic foot ulceration, pressure sores on the foot are prevalent in diabetes. In medical innovation, essential sensing techniques for pressure, temperature, and SpO₂ levels play a pivotal role in preventing complications and advancing care for diabetic patients [5].

In light of this, the IoT solution for foot ulcer prediction that is proposed in this paper is cloud-based. Heat sensors, force-sensitive resistors, and a Max30105 sensor are all components that are incorporated into this system in order to record and evaluate important parameters that are associated with foot health. Real-time data transfer for remote monitoring and analysis is made possible by the incorporation of an ESP32 module, which permits wireless connectivity over Wi-Fi and Bluetooth while also facilitating the transmission of data.



Fig. 1. The Study suggests smart insole warns, prevents diabetes foot ulcers, averting complications.

In Fig. 1, the smart insole system known as SurroSense Rx® was developed by Orpyx Medical Technologies Inc., a Canadian company that specializes in the production of innovative wound care products [6]. This system was implemented by the Manchester Metropolitan team for the purpose of testing. Eight sensors were incorporated into the shoe inserts, and they were designed to continuously record and monitor pressure points along the sole of the foot while the wearer went about their regular activities. An audio-visual vibrating alarm was transmitted to the smartwatch that was accompanied by the device through the use of wireless data transfer. If the smartwatch detected any harmful pressure buildup, it would vibrate to inform the wearer to immediately reduce pressure on the concerned foot zone. The patient may be told to move around to better distribute their weight across both feet, sit down and release pressure from the troublesome foot, or have their shoes examined for foreign items or excessively tight shoelaces [7].

II. PROPOSED SYSTEM METHODOLOGY AND SIMULATION

Diabetic foot ulcers are a common complication of poorly controlled diabetes mellitus. In roughly 15% of diabetic individuals, an open wound or sore will develop on the sole of the foot; this condition is known as a diabetic foot ulcer. A total of 6% of patients with foot ulcers require hospitalization due to complications such as infections. Nontraumatic lower extremity amputations in the United States are most often caused by diabetes, which accounts for 14–24% of all amputations in patients with foot ulcers [8]. In order to avoid amputating a foot, it is necessary to employ a system that can detect foot ulcers early on by monitoring skin and tissue in the legs.

A prototype system for remote sensing and monitoring can't be built without first connecting it to a handful of sensors. The data will be transmitted to the cloud by the controller, which fits with a power backup system. Figure 2 shows a simplified block diagram of the Internet of Things (IoT) smart insole with a sensor that can test skin and leg tissue. The sensors collect data and send it to the Arduino UNO R3 microcontroller, which processes it. After then, the NodeMCU-32 Wi-Fi module sends the data to the cloud so it can be analyzed in depth.

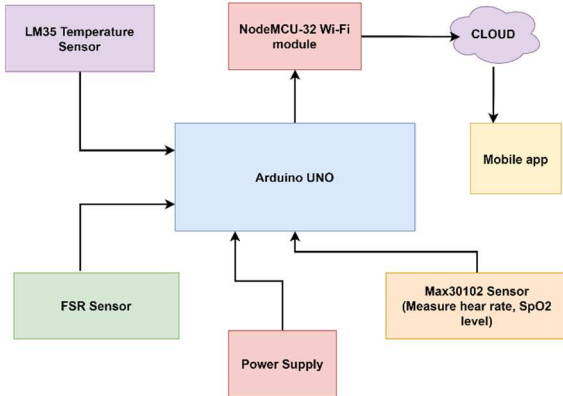


Fig. 2. Foot health monitoring system in block diagram.

The Max30102 sensors can detect pulse oximetry (SpO2) and heart rate (HR) signals [9]. The LM35 Temperature Sensors, which can detect foot temperature, FSRs are used to detect foot pressure so that they can analyze the foot pressure pattern of the patient. A significant portion of this project involves the conversion of analog signals to digital signals.

Foot pressure, temperature, and SpO2 level data are all examples of analogue signals that need to be converted so that they may be used in digital signal processing. The transformation of analogue signals into bits is necessary in order to achieve greater precision [10]. A numerical value will be shown on the cloud server in the form of this digital information, also known as bits. Bits are made up of neither ones nor zeros. Therefore, rather of being stored in bits, it needs to be converted into a decimal or floating-point value. mainly due to the fact that the physicians and their companions will make a diagnosis of the patient based on these data. The Wi-Fi module sends data to the IoT device's server via connecting to the internet [11]. Portrayed graphically. Depending on the facts, the patient's diagnosis may be every day, every week, or even every month. The data is accessible to everyone, everywhere. There won't be any data loss.

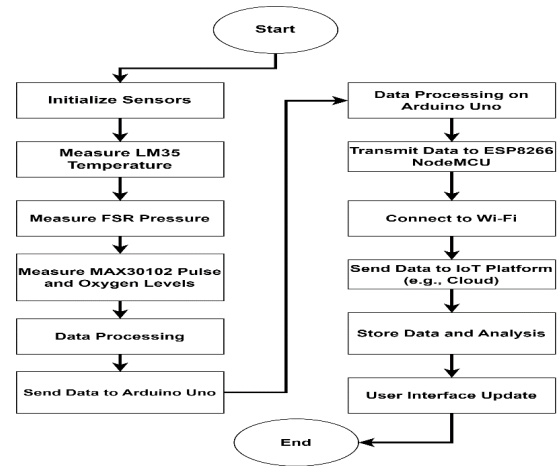


Fig. 3. Integrated Sensor Data Transmission Flowchart.

This flowchart (Fig. 3) illustrates the operation of an IoT-based Smart Insole, incorporating LM35, MAX30102, and FSR sensors. It initiates sensor measurements, processes data on Arduino Uno, transmits information to ESP8266 NodeMCU, and subsequently sends health metrics to an IoT platform for storage and analysis, enabling real-time health monitoring through a user interface.

III. HARDWARE PROTOTYPE DEVELOPMENT FOR TESTING

The proposed circuit utilizes an Arduino microcontroller to process signals from various sensors as shown in Fig. 4, that monitor conditions potentially leading to foot ulcers. So, the Arduino Uno R3 acts as the central processing unit, reading sensor data, processing it, and sending information to a smartphone app via the ESP8266 NodeMCU [12]. So, the NodeMCU provides Wi-Fi connectivity to the system, allowing data transmission to an alert system for monitoring.

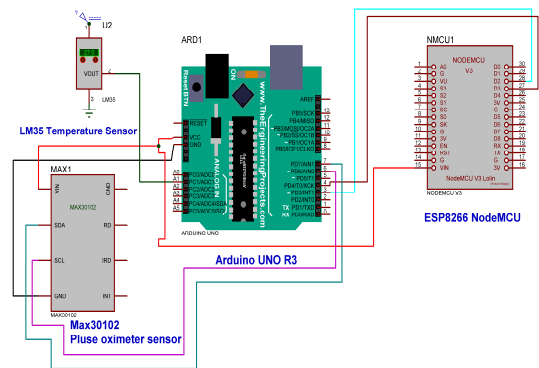


Fig. 4. Developed circuit using proteus.

The LM35 temperature sensor basically measures the ambient temperature in the shoe insole, which is a factor that can affect foot health. force-sensitive resistor, which can also be called an FSR sensor. So, this sensor is placed in the insole to measure the pressure exerted by the foot, and continuous high pressure is a risk factor for ulceration. And last of all, the MAX30102 sensor is used for measuring blood oxygenation (SpO2), which can indicate the overall circulatory health of the foot.

A. System Explanation

FSR Sensor Converts pressure on the foot into a measurable electrical resistance.

$$R_{fsr} = \frac{V_{supply} * R_{divider}}{V_{fsr} - R_{divider}} - R_{divider} \quad (1)$$

R_{fsr} is the resistance of the FSR V_{supply} is the voltage supplied to the circuit. V_{fsr} is the voltage across the FSR. $R_{divider}$ is the resistance of a fixed resistor in series with the FSR [13]. LM35 Sensor outputs a digital signal with the temperature reading. MAX30102 Sensor uses optical sensors to measure the variation in blood volume during a heart pulse to derive heart rate and SpO2 levels.

$$SpO_2 = 104 - 17R \quad (2)$$

where R ranges between ≈ 2 to ≈ 0.23 if SpO2 varies in the range from 70% to 100% [14]. The ESP8266 NodeMCU connects to a Wi-Fi network and transmits data to a web server or a mobile device.

TABLE I. TABLE OF SYSTEM SPECIFICATIONS

Component	Specification	Function
Arduino Uno R3	Microcontroller ATmega328P	Central Processing
ESP8266 NodeMCU	Wi-Fi Module, 80MHz Microprocessor, 4MB Flash	Wireless communication
LM35	Provide data of $\pm 0.5^\circ\text{C}$ and $\pm 0.75^\circ\text{C}$ with a range of -55°C to 150°C	Precise temperature measurement
FSR	Variable resistance	Pressure sensing
Max30102	Heart rate 30-250 bpm, SpO2 70-100%	Heart rate and oximetry

This innovative setup features a smart insole system specifically engineered to monitor conditions, such as excessive pressure, that are known precursors to ulcer formation. The system's core functionality, shown in Table I, hinges on the strategic placement of pulse rate and temperature sensors at the toe area. Once these sensors are securely attached, they embark on a continuous cycle of reading and writing vital data [15]. This data is not only displayed in real-time on a dedicated serial monitor for immediate observation but is also simultaneously transmitted to a specialized cloud server. Within this cloud environment, sophisticated tools, including gauss meters and comprehensive graphical analysis techniques, are employed to meticulously process and interpret the temperature readings. This advanced approach plays a crucial role in the early detection of potential health risks, effectively keeping both users and their healthcare providers informed and alert to any emerging issues that might require intervention.

IV. HARDWARE DEVELOPMENT AND TESTING

Subsequent to obtaining the requisite simulation outcome, the prototype of the Internet of Things-enabled smart insole including a sensor system was constructed in a laboratory setting. Fig. 5 illustrates the circuit subsequent to the installation of all the components. Following the completion of the prototype design and development, performance testing has commenced. The functional testing of the Arduino UNO R3 microcontroller was conducted to verify its capacity to process data from attached sensors. A sequence of test scripts was conducted to emulate sensor data input, assessing the microcontroller's response speed, data precision, and capacity to manage simultaneous data streams [16]. According to the plan, the data will thereafter be transmitted to the user interface via the IoT platform.

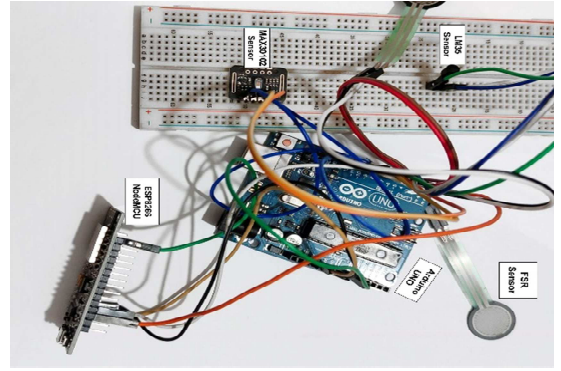


Fig. 5. Prototype hardware development for testing purposes.

After completing the circuit by using the components listed in Table II, the lab testing was executed. The validation process for the Thermal Sensor LM35 focused on verifying temperature measurement accuracy and responsiveness to temperature changes.

TABLE II. TABLE OF SYSTEM COMPONENTS

S/N	Component List	Component Model	Quantity
1	Arduino Uno	Arduino UNO R3	1
2	Thermal Sensor	Thermal Sensor LM35	2
3	Force Sensitive Resistor 0.5 inch	SEN-00049	3
4	Wi-Fi connectivity	ESP32 V1.3 NodeMCU-32	1
5	Measure SpO2 level	Max30102 Sensor	1
6	Others	Bread boards, wires, clips etc.	

Following initial testing of the integrated circuit, the prototype was affixed to a shoe sole and evaluated independently, as illustrated in Fig. 6. The sensor was subjected to fluctuating temperatures in a controlled laboratory setting, and the precision of temperature measurements under diverse settings was evaluated. The evaluation of the FSR pressure sensor entailed the application of regulated pressure at several locations on the insole to precisely map pressure distribution [17]. Collected pressure data was utilized to assess the sensor's precision and spatial resolution in delineating pressure fluctuations.

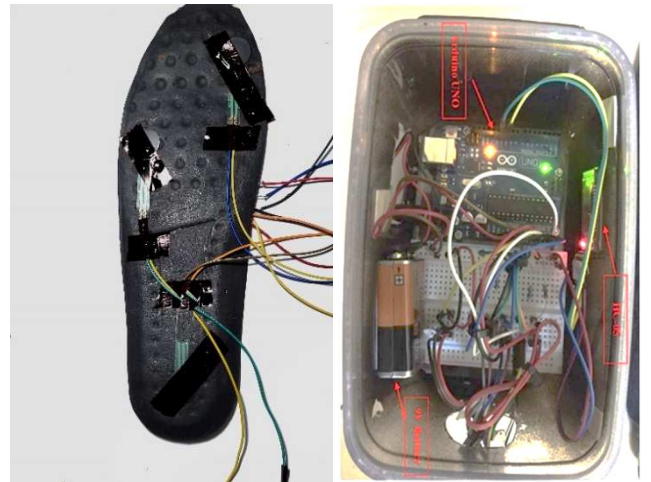


Fig. 6. (A) Parts contained in the box. (B) Integration of the sensors.

The functionality of the ESP32 as a communication bridge between the insole and external storage underwent testing. Simulated data transmission from the insole to an external storage server was performed to evaluate communication reliability, speed, and security features. The Max30102 Sensor's ability to monitor heart rate and oxygen levels accurately was verified through simulated variations in heart rate and oxygen levels [18]. Sensor readings were compared with expected values to evaluate precision and responsiveness to dynamic physiological changes.

A. Hardware Setup

As seen in Fig. 7 below, the circuit and every part of the system aside from the sensors were linked to a breadboard within a box. The Arduino UNO was powered by a 12V battery, and the NodeMCU module esp-32 served as the data transmitter to the cloud.



Fig. 7. Representation of the final Product.

In addition, the insole will be coated to both cushion the user's foot and shield the FSR sensor's surface. The regular insole inside the shoes will then be replaced by the smart insole. The user can then put on the device and use a belt to tighten the blue box around their calf.

B. ThingSpeak setup

ThingSpeak is a Web of Things (IoT) platform that permits users to collect, examine, and visualize information from sensors or devices continuously. To send the information to the ThingSpeak platform, first gather the information utilizing Uno and send the information to the cloud server by means of NodeMCU. To gain permission to the ThingSpeak stage, it needs to make a record on that platform, then, at that point, make another channel, and afterward give the channel name and description. From that point forward, choose a field to show just temperature information from the LM35 sensor. Following saving the channel, there will be various choices, like personal review. In this part, representations, devices, and data can be sent out in JSON, XML, and CSV records. In the Programming interface keys segment, there is a write API key, which is utilized in the coding segment to send the information on the ThingSpeak platform [19].

V. PERFORMANCE ANALYSIS AND DISCUSSION

Real-time sensor data readings that can be shown on a smartphone are the goal of this technology. The NodeMCU ESP32 data communication module, which it employed, was put to the test to see how well it worked in order to accomplish that goal. Testing the synchronization between the ThingSpeak result and the serial monitor in the Arduino IDE revealed that both were displaying the same value at the

same time. Subsequently, the same code was tested to obtain real-time values from all three sensors and showed in Table III. The outcome demonstrates that instantaneous sensor reading is not achievable.

TABLE III. OUTCOMES FROM 90 SECOND STANDING

Time (s)	Temperature (Celsius)	Pressure at Big Toe	Pressure at Meta 1	Pressure at Meta 2	Pressure at Heel
6	30.76	60	82	81	95
12	32.23	78	82	82	95
18	32.23	78	80	81	95
24	32.23	80	81	81	95
30	32.23	76	80	81	95
36	32.23	76	81	81	95
42	32.72	75	81	81	95
48	32.72	71	81	81	95
54	32.72	54	79	80	95
60	32.23	53	79	81	95

Two distinct time delays are used in this performance analysis: one uses 1.5 seconds for each sensor, while the other uses 0.6 seconds. Aside from that, the pressure sensor's maximum thresholds have been established, and the pressure range will show between 0-100 and 90. The red column will display the results for pressures exceeding the maximum level. Therefore, the smartphone will vibrate and notify the user of an impending danger when the pressure displayed is greater than 90 [20].

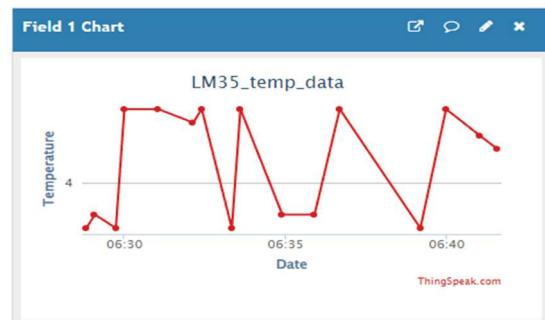


Fig. 8. Visualization of temperature data.

The temperature sensor data readings that are shown on the ThingSpeak platform are shown on Fig. 8. The code performs various calculations to determine the true temperature in the region for the temperature reading.

VI. CONCLUSION

It has been determined that the project's objectives have been successfully accomplished. The purpose of this project is to develop a smart insole that has the potential to limit the activities that its users engage in on a daily basis in order to reduce the risk of developing foot ulcers. Given that the number of diabetes patients continues to rise on an annual basis, this project may be of assistance in providing medical services for the purpose of patient monitoring. The problem can be solved by providing the diabetic patient with the most effective preventative remedy to protect them from developing a foot ulcer through the design of an engineering system that is appropriate. A cloud-based system has been utilized in this study to display the pressure and temperature data from the plantar foot. The user is able to monitor their own health state through the use of their smartphone.

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